

Unified Multimodal Representation Learning: Architectures, Alignment, and Retrieval

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Abstract

A shared embedding space offers a simple interface to heterogeneous data: encode text, images, video, or audio once, then retrieve relevant items by comparing representations. The difficulty is deciding what those representations should preserve. Semantic similarity, exact identity, temporal order, and evidence support impose different requirements, even when the inputs are identical. This review traces the development of multimodal embeddings from paired image–text encoders to modality binding, instruction-conditioned backbones, and omni-modal retrieval. We organize the methods by input coverage, relevance conditioning, representation granularity, alignment strategy, and deployment compatibility. The discussion connects training objectives and data construction to the specific demands of visual documents, long video, speech, environmental sound, and music. Selected quantitative comparisons retain their original benchmark versions, metrics, and retrieval directions, revealing differences that aggregate scores can obscure. Recent studies on sparse–dense outputs, visual identity, interactive queries, and frozen-space expansion further broaden the design space. We conclude that progress toward general-purpose multimodal representations depends on preserving the right evidence for each task while keeping encoding, indexing, and model updates affordable. Relation-specific evaluation, capability retention, and tests against deployed indexes are central to that goal.

Keywords: multimodal embeddings; contrastive learning; universal retrieval; vision–language models; audio–video representation; dense retrieval.

1 Introduction

An embedding system turns an input into a representation that can be stored, compared, and reused. For text search, the input may be a query or passage; for multimodal search it may instead be an image, a document page, a video segment, an audio recording, or a combination of these. The practical appeal is substantial: expensive encoding can be performed before a query arrives, while a comparatively cheap similarity operation supports retrieval over a large collection. Shared representations also make it

possible to search one modality using another, such as finding a video with a textual description or retrieving an illustrated report from a photographed diagram.

CLIP established a particularly influential recipe for this setting: learn image and text encoders from paired examples and compare their normalized representations [1]. ALIGN demonstrated a related direction using noisy image–text supervision at scale [2]. These models showed that language can provide a flexible interface to visual concepts. However, the ability

to match an image to a caption does not automatically imply strong passage retrieval, visual identity discrimination, temporal reasoning, or acoustic understanding. The same vector can preserve the topic of a scene while discarding the specific object, temporal order, or utterance that a search task requires.

Subsequent work expanded the set of connected modalities. ImageBind uses an image-centered binding strategy, while LanguageBind emphasizes language-based semantic alignment [3, 4]. In parallel, instruction-tuned text embedding research demonstrated that retrieval depends on the requested task, not only the literal content of the query [5, 6]. Multimodal language models then offered a route to conditional representations of heterogeneous inputs, as illustrated by E5-V, VLM2Vec, MM-Embed, and GME [7–10]. More recent systems extend this route to video, visual documents, and audio [11–14].

As these approaches converge, the meaning of universal becomes less precise. Accepting several input types says little about which relevance relations a model can recover. Strong benchmark coverage may come with expensive preprocessing, and a new modality can improve the average while weakening an established text index. Specialized encoders also remain competitive on tasks whose local or temporal evidence is poorly represented by a general-purpose vector. These distinctions motivate the organization of this review.

The review follows three questions: how are modalities connected, what information survives in the representation, and how is that information used at retrieval time? We first establish a taxonomy of architectures and alignment strategies, then examine document, video, and audio retrieval in detail. The benchmark discussion separates comparable measurements from results obtained under different protocols. We close with the indexing and evaluation problems that arise when a shared space must serve several tasks over time. Figure 1 places the main developments in context.

1.1 Scope and relation to earlier surveys

The taxonomy of Baltrušaitis et al. organizes multimodal machine learning around representation, translation, alignment, fusion, and co-learning [15]. Manzoor et al. examine the evolution of multimodal representation learning and its applications [16]. Later surveys address embeddings in the foundation-model era and representation alignment more broadly [17, 18]. Visual document retrieval also has a rapidly developing survey literature [19]. These perspectives provide valuable context. Our narrower unit of analysis is the reusable retrieval representation: what is encoded offline, what is conditioned on a query, how candidates are scored, and what evidence supports cross-task and cross-modal reuse.

We include text-only retrieval foundations when they explain the design of multimodal embeddings, but do not attempt to survey all language modeling, audiovisual generation, or multimodal reasoning. A generative model is relevant here when its features, reasoning traces, or supervision are used to construct retrievable representations. A foundation model that accepts images and audio is not classified as an omni-modal embedder solely on that basis. Similarly, masking and any-to-any prediction are related representation objectives, but they do not by themselves establish a common cosine space for retrieval.

1.2 Literature selection and evidence

We selected papers through citation tracing and targeted searches on vision–language alignment, modality binding, universal retrieval, visual documents, video, language–audio learning, and embedding geometry. The emphasis is on architectural transitions and retrieval evaluation. Text-only work is included where it explains an objective, representation interface, or indexing method used by multimodal systems.

References are tied to primary records and identifiable archive versions. The numerical comparisons in Section 10 are transcribed from the

Evolution of unified multimodal representations

Selected methods, design families, and references in this review

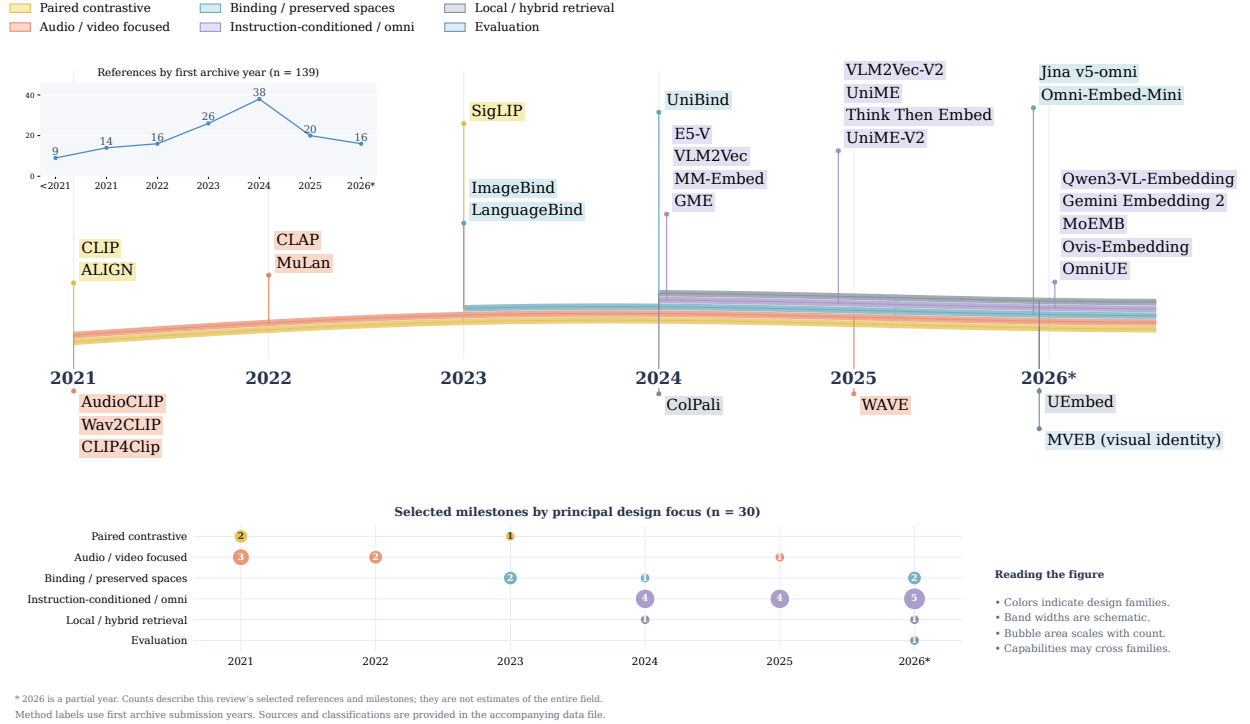


Figure 1: Selected milestones in multimodal representation learning, grouped by first archive submission and principal design focus. The upper-left panel counts the 139 references in this review; bubble areas count the 30 selected milestones. Colored ribbons identify design families schematically and do not measure prevalence or performance. The 2026 counts are partial. Methods may span several families; the grouping does not imply architectural ancestry.

cited reports; they are not new experiments. We keep each report’s benchmark definition and distinguish proposed diagnostics from measured results. All diagrams summarize methods or design choices rather than show empirical performance.

2 Problem formulation and dimensions of universality

2.1 Inputs, relevance, and shared spaces

Let $\mathcal{M}$ be a set of modalities and let x denote a possibly structured input. A document can contain text, page images, and layout; a video can contain sampled frames, audio, timestamps, and subtitles. We write $x = (x^{(m)})_{m \in \mathcal{M}_x}$, where

$\mathcal{M}_x \subseteq \mathcal{M}$. A dense encoder produces

$$\mathbf{z} = f_{\theta}(x, t, r) \in \mathbb{R}^d, \quad \hat{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2} \quad (1)$$

where t is a task instruction and r denotes a role, such as query or candidate. The common cosine score is $s(q, c; t) = \widehat{f_{\theta}(q, t, \text{query})}^{\top} \widehat{f_{\theta}(c, t, \text{candidate})}$. Distinct towers may implement f_{θ} for different modalities; the notation does not require shared parameters.

Relevance is an external relation $R_t(q, c)$, rather than an intrinsic consequence of closeness in the representation space. “Find the same product,” “find a visually similar product,” and “find an image illustrating this concept” define different relations. Even within the text domain, topical similarity, evidence retrieval, paraphrase matching, and instruction compliance are different tasks. A shared space is successful when

its induced ranking approximates the specified relation under the data distribution of interest.

Query conditioning and candidate conditioning have different systems consequences. If a task instruction changes only the query representation, one candidate index can support several search intents. If candidate representations also depend on the instruction, a system may need multiple indexes or online re-encoding. Many benchmark settings fix instructions in advance, masking this distinction. We therefore treat the persistence of candidate representations across tasks as part of operational universality.

2.2 Dense, sparse, and multi-vector representations

A dense single-vector model summarizes an input into one compact vector. A learned sparse model instead outputs weights over a vocabulary or feature inventory. A multi-vector model preserves a set of representations $Z_x = \{\mathbf{z}_{x,j}\}_{j=1}^{L_x}$, often associated with tokens or patches. ColBERT introduced an influential late-interaction approach, followed by compression and retrieval-engine work in ColBERTv2 and PLAID [20–22]. A common score is

$$s_{\text{late}}(q, c) = \sum_{i=1}^{L_q} \max_{j \leq L_c} \hat{\mathbf{z}}_{q,i}^\top \hat{\mathbf{z}}_{c,j}. \quad (2)$$

This retains local matching information that pooling may erase, but requires specialized indexing and additional scoring work. ColPali applies the general idea to document page images [23]. Sparse grounded image–text representations appear in STAIR, while UEmbed extends unified sparse and dense outputs to decoder-based multimodal embedding [24, 25]. These approaches show that a universal system need not expose only one dense vector.

Figure 2 compares the representation and index interfaces. Representation granularity should be chosen with the retrieval unit. A page-level embedding is appropriate for finding a page, but insufficient by itself for returning a precise supporting region. A clip-level vector can retrieve an event while failing to identify its

time interval. A whole-recording vector may preserve the speaker or setting without preserving a short target sound. Retrieval and localization are related but separate requirements; the index unit can be smaller than the user-visible artifact, or hierarchical retrieval can connect the two.

2.3 An operational taxonomy

Table 1 gives five axes for describing an embedding system. We propose these axes as an analytical framework rather than a benchmark derived from a particular source. Each axis contains several decisions, and the axes interact. For example, adding audio may expand coverage while increasing query latency; preserving local representations may improve document grounding while increasing storage.

A useful capability profile can be written as $\mathcal{U} = (C_{\text{input}}, C_{\text{task}}, C_{\text{granularity}}, C_{\text{transfer}}, C_{\text{operation}})$. These components need not be collapsed into a scalar. Their weights depend on the application: a model with stronger audio retrieval may be less useful than one that preserves a compact, reusable text index. This view also makes absence meaningful: “not evaluated” is different from “unsupported,” and both differ from a measured failure.

3 From paired alignment to general-purpose visual representations

3.1 The CLIP recipe and its assumptions

For a batch of paired image and text representations $(\mathbf{v}_i, \mathbf{u}_i)$, a directional contrastive loss is

$$\mathcal{L}_{v \rightarrow u} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\mathbf{v}_i, \mathbf{u}_i)/\tau)}{\sum_{j=1}^B \exp(s(\mathbf{v}_i, \mathbf{u}_j)/\tau)}, \quad (3)$$

with temperature τ and a corresponding text-to-image term. The symmetric form rewards matching pairs and separates them from other batch examples. Contrastive predictive coding

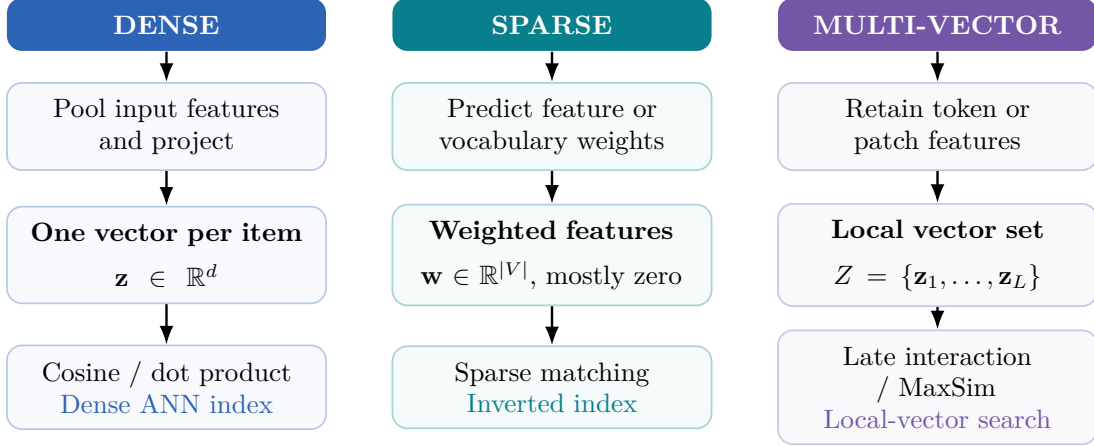


Figure 2: Representation choices determine the retrieval interface. Dense pooling compresses evidence into one vector, sparse outputs expose weighted features, and local vectors defer interaction until scoring. ColBERT and ColPali use local interaction; STAIR and UEmbed explore sparse or sparse–dense representations [20, 23–25].

Table 1: Five axes for assessing a multimodal embedding system. Coverage should be specified at the task and retrieval-unit level.

Axis	Principal questions	Evidence needed
Input coverage	Which modalities and mixed inputs are accepted? What survives preprocessing?	Explicit input formats, lengths, frame/audio limits, missing-modality tests.
Relevance conditioning	Which relation is learned? Can instructions or interactions change it?	Task-specific and instruction-contrast evaluations.
Interaction granularity	One vector, sparse weights, or local vectors? Can the result be localized?	Retrieval and grounding metrics at appropriate units.
Alignment strategy	Joint learning, a frozen anchor, distillation, or space composition?	Within-modal retention and cross-modal transfer tests.
Operational compatibility	Can one index be reused? At what latency, memory, and versioning cost?	Measured encoding, indexing, search, and migration costs.

provides an earlier foundation for this family of objectives [26]. In CLIP-style learning, the paired caption is a convenient supervisory signal, but it is not a complete specification of the image’s contents [1]. A caption may omit a second object, a subtle spatial relation, or a distinguishing mark. The representation is therefore shaped by what the pairing process makes recoverable.

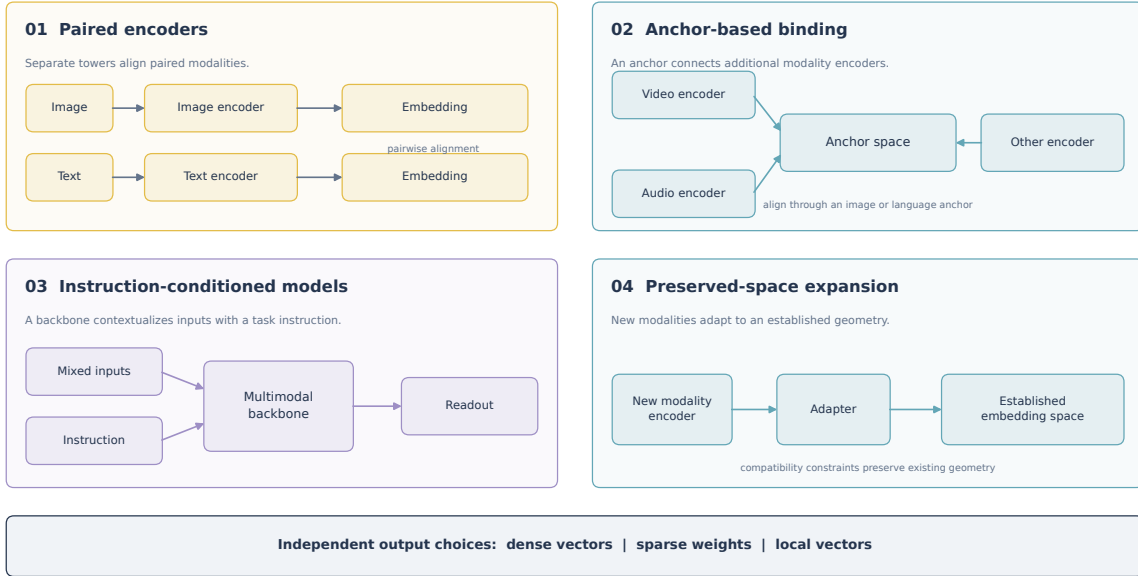
This observation explains why scale and data quality are inseparable. More pairs can improve coverage, yet redundant or underspecified captions can reinforce shortcuts. LAION and DataComp make data construction and selection central objects of study [27–29]. Work on CLIP

data and filtering further examines how training distributions affect downstream performance [30, 31]. A large corpus does not automatically provide the relation-specific negatives required for fine-grained search.

SigLIP replaces the batch-normalized contrastive formulation with a pairwise sigmoid objective [32]. This changes the relationship between pairs, normalization, and optimization, while retaining the broader aim of aligning images and text. SigLIP 2 broadens the recipe toward multilingual understanding, localization, and dense features [33]. Such extensions are relevant to universal embeddings because the

Architectural routes to multimodal representations

Four design patterns; topology illustrates information flow rather than a ranking of methods.



Coverage, parameter sharing, and space compatibility are separate choices; these patterns can overlap.

Figure 3: Four architectural routes to reusable multimodal representations. Parameter sharing, modality coverage, and metric-space compatibility are separate design choices. Each route can be combined with different output granularities.

quality of the visual encoder constrains what later adapters can preserve. An adapter cannot reliably recover a detail that its input features have already discarded.

3.2 Global semantics versus local correspondence

A global image vector is attractive because it is cheap to store and compare. Its weakness is the compression of many possible relevance relations into one representation. FILIP uses fine-grained interaction between image and text features, and RegionCLIP emphasizes region-level language-image correspondence [34, 35]. Alpha-CLIP conditions visual representation on a selected region [36]. These directions motivate a broader principle: local evidence should enter the representation or the scoring function when users ask about local content.

Pooling and interaction occupy different

points on an efficiency continuum. Pooling moves computation into offline encoding but fixes what information remains available at search time. Late interaction preserves local evidence but increases storage and scoring complexity. Cross-encoding a query with each candidate offers the richest direct interaction but is usually restricted to a shortlist. A retrieval system can combine all three: pooled first-stage search, local-vector refinement, and a final cross-modal reranker. The resulting system should be evaluated as a pipeline, because a strong reranker cannot recover a relevant candidate excluded by first-stage retrieval.

3.3 Long descriptions, recaptioning, and language capacity

Short captions can underdescribe scenes, and fixed text limits can truncate useful constraints. Dense-caption evaluation, Long-CLIP, DreamLIP, and LoTLIP investigate longer or richer

language supervision [37–40]. VeCLIP and large-scale recaptioning use language models to enrich supervision [41, 42]. LLM2CLIP further connects stronger language representations to cross-modal learning [43]. These works should not be interpreted as showing that more text is invariably better. Additional words help when they describe discriminative visual evidence; they can harm when they add hallucinated objects or relations.

Dense captions also change the effective training task. A concise caption may reward broad object matching, while a detailed caption introduces attributes, counts, and relations. If all details are pooled into one target, the model still needs a mechanism for distinguishing which detail a query requests. Richer supervision and instruction conditioning are therefore complementary. A useful evaluation should contrast short and long queries, check truncated versus retained constraints, and separate errors caused by token limits from errors caused by compositional reasoning.

Language capacity matters beyond English. Chinese CLIP, AltCLIP, multilingual Jina CLIP, and Meta CLIP 2 demonstrate distinct routes to expanding language coverage [44–47]. A multilingual image space is not established by averaging across language-specific retrieval scores if scripts, query styles, or domain vocabulary differ substantially. Multilingual evaluation should ask whether the same image remains retrievable across paraphrases and languages, and whether cross-language text retrieval survives visual training.

3.4 Transfer, freezing, and efficient visual backbones

LiT studies locked-image text tuning, providing an early example of preserving one strong encoder while learning its interface to another modality [48]. BLIP and BLIP-2 connect representation learning to captioning and frozen-model adaptation, while CoCa combines contrastive and captioning objectives [49–51]. These are relevant precedents for later systems that

borrow a multimodal language model’s features but optimize a separate embedding objective.

Efficiency-oriented methods such as TinyCLIP and MobileCLIP explore distillation and training recipes for smaller image–text models [52, 53]. Perception Encoder examines where useful visual embeddings occur inside a network [54]. Together, these directions caution against identifying the largest final hidden state with the best retrieval representation. Backbone size, feature layer, pooling, and projection should be treated as independently testable choices. For high-volume retrieval, a smaller encoder with reliable domain coverage may yield a better system-level tradeoff than a larger encoder whose incremental ranking gains require substantially more preprocessing.

4 Binding modalities and preserving representation geometry

4.1 Anchors reduce supervision requirements

Directly paired training examples become harder to obtain as the number of modalities grows. An image may coincide with audio, depth, thermal information, or inertial measurements, but all modalities need not appear in every example. ImageBind addresses this situation through image-centered alignment [3]. LanguageBind uses language as a semantic alignment interface for multiple modalities [4]. These approaches replace a requirement for every modality pair with a network of alignments through an anchor.

The advantage is organizational as well as statistical: separate modality datasets can contribute to a common representation objective. The limitation is that the anchor mediates what information is shared. Language often omits low-level acoustic texture or exact visual identity, while image semantics may inadequately specify spoken content. Alignment to an anchor can thus produce useful semantic transfer without preserving all within-modal distinctions.

This is a predictable consequence of the information present in paired examples, not simply a failure of model size.

Shared-space transfer also lacks automatic transitivity. If audio aligns well with images and text aligns well with images, audio-text retrieval may improve, but its calibration need not match either directly supervised relation. The relevant concepts, noise levels, and modality distributions can differ. Evaluation should include untrained pairings and within-modal retrieval, rather than infer complete all-to-all compatibility from a few aligned pairs.

4.2 Expansion, composition, and balanced spaces

ONE-PEACE and Meta-Transformer investigate broader multimodal representation frameworks [55, 56]. Point-Bind extends alignment toward point clouds, showing that the design space reaches beyond the four modalities emphasized in this survey [57]. UniBind, FreeBind, and the two distinct works named OmniBind explore augmentation, knowledge fusion, or the composition of pretrained spaces [58–61]. Their shared challenge is making representations useful across modalities whose datasets and encoders have unequal coverage.

Space composition should be assessed against the possibility of negative transfer. If a transformation improves an underrepresented modality but shifts established image-text neighborhoods, an aggregate score can conceal regressions. Balanced evaluation should measure both the new relation and retention on existing ones. It should also distinguish alignment quality from the strength of the constituent encoders; a stronger audio tower can improve retrieval even if the binding mechanism is unchanged.

Massively multimodal masked models such as 4M and 4M-21 broaden representation learning through tokenized modalities and prediction tasks [62, 63]. They are related to binding, but their ability to perform many conditional predictions should not be equated with evidence that a single retrieval vector works for all those tasks.

The distinction is useful: shared token interfaces, shared backbone parameters, and shared metric geometry are three different forms of unification.

4.3 Alignment does not mean identical distributions

Wang and Isola analyze contrastive learning through alignment and uniformity, while Mind the Gap examines separation between modality distributions [64, 65]. A common ambient vector space can contain modality-specific clusters or offsets. High cross-modal retrieval accuracy therefore does not require complete distributional coincidence, and visual overlap in a two-dimensional projection does not establish useful alignment.

For an operational retrieval system, the key issue is whether the geometry preserves the desired rankings and whether scores remain usable across candidate types. If a mixed index contains text, images, and audio, modality-dependent score distributions may favor one candidate type even when within-modality rankings are strong. This suggests evaluating a genuinely mixed candidate pool, calibrating scores on held-out tasks, and reporting candidate-type exposure. Separate homogeneous benchmarks cannot fully reveal this failure mode.

A simple retention diagnostic compares embeddings before and after extension. Let f_0 be an original text encoder and f_1 the extended system. Pairwise score drift can be summarized as

$$D_{\text{score}} = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{text}}} |s_{f_1}(x,y) - s_{f_0}(x,y)|. \quad (4)$$

This diagnostic complements task evaluation. Small score drift can coexist with ranking changes near a decision boundary, and a coordinate rotation can preserve pairwise scores while breaking compatibility with an old stored index. Consequently, score retention, ranking retention, and coordinate compatibility require separate checks.

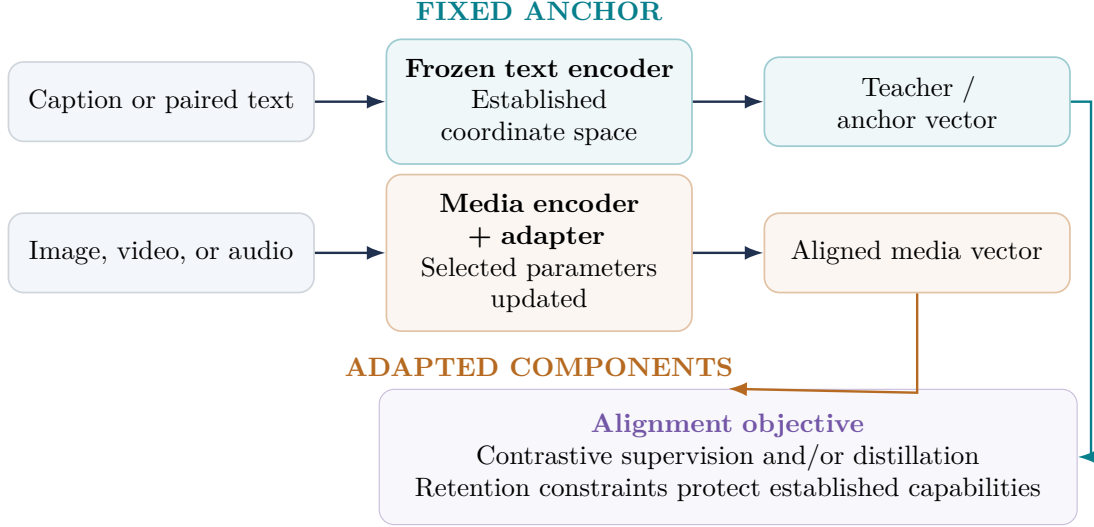


Figure 4: Expanding a frozen embedding space. Text supplies an anchor while selected media components learn compatible outputs. Jina v5-omni and Omni-Embed-Mini instantiate different versions of this preservation-oriented strategy [13, 66]. Teal indicates fixed components; amber indicates components selected for adaptation.

4.4 Frozen spaces as a deployment constraint

Geometry-preserving extension treats an established text space as an asset that should survive the addition of modalities. Figure 4 illustrates this training pattern. Jina embeddings v5-omni uses locked aligned towers and trains connectors around a frozen text foundation [13]. Omni-Embed-Mini similarly keeps text-side parameters fixed and uses dense-caption embeddings from the frozen text backbone as teacher targets for modality expansion [66]. These approaches make backward compatibility a first-class training consideration.

Freezing text weights offers a precise guarantee at the parameter level, but end-to-end compatibility also depends on tokenization, preprocessing, instruction templates, pooling, normalization, and numerical implementation. Even a coordinate-compatible text encoder does not guarantee that new media vectors will be well calibrated in an existing mixed index. The correct deployment question is therefore twofold: whether established text representations remain stable, and whether the added modalities retrieve the intended candidates without disrupt-

ing exposure or score calibration.

5 Multimodal language models as embedding backbones

5.1 From generation to reusable representations

A decoder trained to generate text already computes contextual features, but its generative objective does not ensure that a pooled feature is suitable for nearest-neighbor retrieval. Text embedding methods such as LLM2Vec, NV-Embed, and generative representational instruction tuning examine ways to adapt language-model representations [67–69]. Their relevance to multimodal embedding is architectural: a model’s capacity to interpret an input must be connected to a training objective that shapes distances between examples.

E5-V explores universal embeddings using a multimodal language model, while VLM2Vec trains vision–language backbones for many embedding tasks [7, 8]. MM-Embed and GME extend the universal retrieval direction with task and data design tailored to heterogeneous query–

candidate relations [9, 10]. These models differ from conventional paired towers because an input can contain both language and media, and a task instruction can participate in contextualization before pooling.

Several representation choices remain consequential: which layer is read, whether a dedicated token is used, how padding is handled, whether the candidate uses the same instruction as the query, and how variable media tokens are pooled. Generative competence should not be used as a proxy for these choices. A model can explain why an image is relevant while producing an embedding that fails to place it near the query. Conversely, a compact encoder can retrieve it correctly without generating an explanation.

5.2 Instruction conditioning and mixed queries

Instruction-conditioned embeddings aim to encode the requested relation along with the query content. INSTRUCTOR illustrates this principle in text, while FollowIR and Promptriever highlight the difficulty of evaluating instruction-sensitive retrieval [5, 6, 70]. In a multimodal query, the instruction may specify how to use an accompanying image: search for the same item, an alternative view, a different color, or a semantically related scene.

UniIR supplies a training and benchmarking perspective on universal multimodal information retrieval, while MagicLens focuses on open-ended instructions for image retrieval [71, 72]. VISTA and MARVEL investigate ways to connect visual inputs to established retrieval representations [73, 74]. The broader lesson is that modality fusion should be relation-aware. Simply adding image and text vectors can produce a topic mixture while ignoring whether the text modifies, excludes, or selects part of the image.

Instruction compliance needs contrastive evaluation. If two prompts request different relations for the same input, the expected rankings should differ in an interpretable way. Comparing average retrieval with one fixed prompt cannot establish this property. We recommend controlled

instruction pairs, including positive versus negative constraints and semantic versus identity queries, with the candidate set held constant. This design exposes prompt insensitivity and tests whether the model uses the instruction rather than relying on a familiar input template.

5.3 Synthetic supervision and negative construction

Universal retrieval datasets are often assembled from heterogeneous tasks rather than a single coherent pairing process. MegaPairs and mmE5 investigate synthetic data construction for multimodal and multilingual retrieval [75, 76]. LLaVE introduces hardness-weighted contrastive learning, while UniME and UniME-V2 explore universal embedding learning and model-assisted judgment [77–79]. The appeal of synthetic data is the ability to construct explicit instructions and relations at scale. The risk is that a teacher’s visual omissions or language habits become systematic supervision biases.

Hard negatives are especially important when basic topical matching has become easy. Text retrieval work on guided negative selection and NV-Retriever motivates careful mining rather than treating every other batch example as equally informative [80, 81]. In multimodal data, a near neighbor may be a valid alternative positive: two images can show the same event, or two videos can independently satisfy a broad query. Mining should therefore be linked to the intended relation, with a mechanism to identify ambiguous or false negatives.

Let P_i denote the set of accepted positives for query q_i . A multi-positive contrastive numerator can sum over P_i instead of treating one item as the sole correct match. This reduces an artificial penalty on equivalent answers, but its usefulness depends on the quality of positive-set annotation. For identity retrieval, positives may share an instance; for evidence retrieval, they may independently support the claim; for semantic retrieval, they may express the same concept. These sets are not interchangeable.

5.4 Reasoning before or during embedding

Think Then Embed and reasoning-guided embedding methods examine whether intermediate generative context can improve representations [82, 83]. Embed-RL investigates reinforcement learning for reasoning-driven multimodal embeddings [84]. These methods introduce a new distinction between offline and query-time reasoning. Generating richer descriptions of candidates before indexing amortizes cost over future searches, while query-time reasoning increases latency for every request.

Reasoning can make implicit constraints explicit, but it can also replace raw evidence with an incorrect explanation. A caption-based representation of a chart may miss a number; a generated summary of a video may reverse event order. Retrieval evaluation should consequently preserve access to the original modality and test whether generated context improves difficult relations without harming exact details. A generated rationale is not evidence of a correct retrieval decision unless its content is checked against the input.

Distilling a reasoning teacher into a faster embedder offers another tradeoff. The teacher can supply ranking or representation targets while the student retains a fixed-cost inference path. The relevant comparison is then not merely student versus baseline accuracy: it includes teacher generation cost, data quality, student query latency, and whether teacher-specific errors transfer. This is particularly important when the student is expected to serve domains that the teacher did not adequately cover.

5.5 Recent architectural extensions

Qwen3-VL-Embedding pairs embedding models with a multimodal reranking framework, emphasizing that retrieval and ranking are complementary stages [85]. WeMM-Embedding and Ovis-Embedding provide further technical reports on scaling unified multimodal representations [86, 87]. MoEMB explores mixture-of-experts capacity for universal embeddings [88].

Active parameters, total parameters, and media-token count should be reported separately for such systems; a smaller active-parameter count does not imply a proportionally smaller memory footprint or lower end-to-end latency.

Recent work also widens the output and query interfaces. UEmbed produces sparse and dense representations from a decoder-based model [25]. OmniUE uses user-selected visual regions and audio spans in omni-interactive queries [89]. MVEB brings visual identity discrimination into universal embedding evaluation [90]. These developments expose capabilities that broad semantic benchmark averages only partially measure. A unified backbone may need several representation heads or scoring mechanisms to support them reliably.

6 Visual documents: retrieval beyond object semantics

6.1 Document pages are multimodal retrieval units

A document page combines language, layout, typography, figures, and spatial relations. OCR-based retrieval extracts text and then treats the page as a text object; this is efficient and often useful, but can lose chart structure, equations, and the association between labels and visual regions. Screenshot-based embedding preserves these signals at the visual input level [92]. The decision to encode a page image is therefore not only an alternative preprocessing route: it changes what information the retrieval model can access.

ColPali uses visual features and late interaction for document retrieval [23]. VisRAG connects vision-based retrieval to retrieval-augmented generation on multimodal documents [93]. MMDocIR benchmarks multimodal retrieval for long documents, while ViDoRe V2 advances visual retrieval evaluation [94, 95]. The important distinction is between page discovery and evidence extraction. A retrieved page may be topically relevant but still contain several conflicting values; a generator needs the region

Table 2: Representative architectural routes. The examples illustrate design choices; the table is not a ranking or a claim that every listed model supports every modality.

Route	Representative works	Main design tension
Paired contrastive towers	CLIP, ALIGN, SigLIP [1, 2, 32]	Efficient matching versus local and relation-specific detail.
Anchor-based modality binding	ImageBind, LanguageBind [3, 4]	Transfer without all-pair supervision versus anchor information loss.
Instruction-conditioned MLLM	E5-V, VLM2Vec, GME [7, 8, 10]	Flexible mixed inputs versus encoding cost and prompt dependence.
Visual late interaction	ColPali, PreFLMR [23, 91]	Local evidence retention versus index and scoring complexity.
Omni-modal backbone	WAVE, Gemini Embedding 2 [12, 14]	Broader modality coverage versus imbalance and input budgets.
Preserved anchor geometry	Jina v5-omni, Omni-Embed-Mini [13, 66]	Existing-space retention versus limits of teacher and connectors.
Unified output interfaces	STAIR, UEmbed [24, 25]	Dense semantics and sparse matching versus calibration and maintenance.

or table cell that supports its answer.

6.2 Text extraction and visual encoding are complementary

OCR and visual encoding have different failure modes. OCR can mishandle unusual scripts, rotated text, or complex tables; visual encoding can struggle with small characters or exact numerical distinctions. A hybrid system can retrieve through both text and image representations, then merge and rerank candidates. The merge should use a calibrated or rank-based method rather than assume raw scores from different encoders are comparable.

PreFLMR emphasizes fine-grained late-interaction retrieval across modalities [91]. Such local matching is particularly relevant when a query contains a named entity, a rare symbol, or a specific numerical constraint. Dense page vectors can complement it by identifying semantically relevant pages even when exact vocabulary differs. The representation should match the intended workflow: finding a report, locating a page, selecting a region, and grounding an answer are distinct stages with different candidate pools.

Document preprocessing must also be re-

ported. Resolution, crop policy, page segmentation, OCR inclusion, and maximum token count can materially change the information presented to the model. An apparent architecture advantage may partly reflect a higher-resolution input budget. Benchmark comparisons should disclose these budgets, particularly when large documents must be resized or truncated. A model that accepts long inputs can still fail if the visual encoder removes the small detail on which the answer depends.

6.3 Long-context retrieval and hierarchical indexing

LongEmbed studies long-context text retrieval, and MMLongEmbed extends evaluation to multimodal long-context scenarios [98, 99]. The central issue is not only the maximum accepted length, but the retention of a small relevant item amid unrelated material. Whole-document pooling may dilute a rare detail even when every token fits within the context limit.

Figure 5 connects retrieval units across modalities. A hierarchical index can encode documents, pages, and regions separately. Coarse retrieval narrows the search to likely documents; finer retrieval identifies evidence within them. This

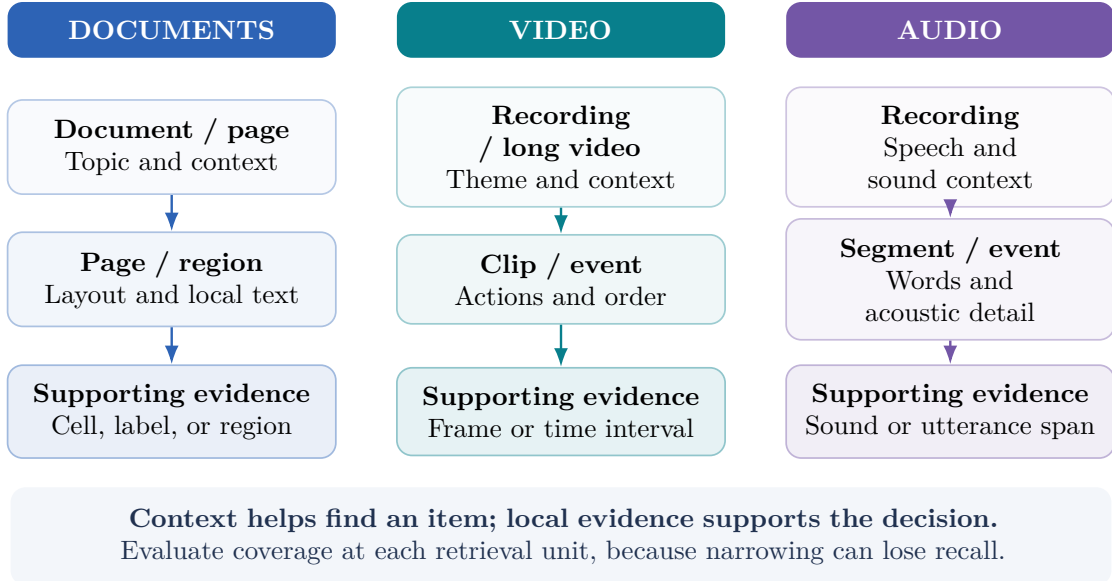


Figure 5: Retrieval units across modalities. Coarse representations capture context, while finer units retain the evidence required by a query. MMDocIR, LoVR, and MSEB illustrate the need for document, temporal, and acoustic evaluation at appropriate granularities [94, 96, 97].

introduces a recall bottleneck at each level. A document-level vector that misses the relevant report prevents downstream page search from recovering it, so systems should evaluate both hierarchical recall and the final evidence quality. An alternative is a flat page index with document-level aggregation, trading more entries for fewer coarse-stage errors.

Cross-page questions require special treatment. If evidence spans two pages, retrieving only one cannot fully support the answer. Evaluation can annotate sets of supporting pages and measure evidence-set coverage, rather than only a single relevant page. This proposed diagnostic also applies to audiovisual evidence spread across segments. It connects long-context embedding to the practical need to assemble sufficient evidence, not merely a topically plausible item.

7 Video embeddings: temporal structure and audiovisual context

7.1 From image encoders to temporal representations

Early video-text retrieval routes adapt image-text representations or jointly train image and video features, as illustrated by Frozen in Time and CLIP4Clip [100, 101]. VideoCLIP and X-CLIP explore temporal and multi-grained contrastive learning [102, 103]. Their common challenge is that a video is not simply a larger image. Frame order, motion, event duration, and transitions can determine relevance.

Temporal pooling is an efficient baseline, but any order-invariant pooling function maps a frame sequence and its permutation to the same aggregate if the constituent frame features are unchanged. It therefore cannot distinguish “entering before sitting” from “sitting before entering” by order alone. A temporal model or positional representation can in principle preserve this difference, but training must make it relevant. Caption supervision that names

only the objects may reward an order-insensitive shortcut.

InternVideo, InternVideo2, VideoPrism, and Unmasked Teacher illustrate broader video representation and scaling directions [104–107]. InternVid contributes video–text data at scale [108]. These foundations can improve the features used by a retrieval model, but recognition or generative scores should not be substituted for retrieval evidence. A representation’s usefulness depends on the query relation and the candidate granularity.

7.2 Sampling policies are part of the model

Sampling determines which events are visible to an encoder. Uniform sampling is reproducible and simple, but can miss a short event. Adaptive sampling may focus on change points or salient frames, but can bias the representation toward the sampling model’s notion of importance. Audio and subtitles add complementary signals, although they can also introduce shortcuts when the intended task is visual retrieval.

A fair video comparison should report frame count, temporal coverage, resolution, clip duration, subtitle use, and audio inclusion. The same backbone can receive substantially different effective evidence under different policies. Query-independent preprocessing is attractive because it can be performed once, but may omit the moment that a future query requests. Query-dependent resampling can help but changes the offline retrieval assumption and introduces additional online work.

We propose reporting an evidence visibility rate alongside retrieval accuracy for controlled temporal tests: the proportion of examples in which the relevant event is included in the sampled input. If visibility is low, an encoder is being penalized for missing evidence it never received. If visibility is high but order-sensitive retrieval remains poor, the failure is more directly attributable to representation or training. This separation makes temporal improvements easier to interpret.

7.3 Unified video and audiovisual retrieval

VLM2Vec-V2 expands multimodal embedding toward videos, images, and visual documents [11]. WAVE studies unified audio–visual embeddings with a multimodal language-model backbone [12]. Universal video retrieval through a synthesized multimodal pyramid curriculum explores the relation between supervision at different temporal scales and generalization [109]. These methods connect video retrieval to the universal embedding problem: text-to-video matching is only one of several possible relations.

Audiovisual fusion requires deciding which stream carries the evidence. A barking sound may be relevant even when the animal is outside the frame; a subtitle may name an object absent from the scene. Treating every synchronized pair as semantically equivalent can blur such distinctions. Evaluation should therefore include modality ablation, missing-stream tests, and queries whose evidence is confined to one stream. A fused model should also be tested on mismatched audio and video to detect whether it follows the more salient but irrelevant modality.

Long-video retrieval adds a further hierarchy. LoVR distinguishes retrieval needs that are not captured by a short-clip benchmark [96]. A theme can recur across a long recording, while an exact clip occupies a narrow interval. The index can store both global and local representations, with a scoring policy that depends on the query. A global summary is appropriate for a broad theme; temporal localization is required for a specific event. Combining the scores without defining the intended relation risks hiding the difference.

8 Audio and music: semantic binding is not speech recognition

8.1 Routes to language–audio alignment

AudioCLIP and Wav2CLIP extend CLIP-related representations to audio through different alignment strategies [110, 111]. CLAP and subsequent large-scale language–audio pretraining learn acoustic concepts from text supervision [112, 113]. Natural-language supervision for general-purpose audio representations broadens the motivation beyond a single task [114]. These methods enable queries such as finding a particular environmental sound, rather than requiring a classifier label inventory.

Speech, environmental sound, and music nevertheless expose different relevance relations. A spoken sentence contains linguistic content, speaker identity, prosody, and background sound. Automatic speech recognition preserves a useful approximation to the words, but often removes the other signals. Native audio embedding can preserve some of them, while its semantic alignment may still favor the transcript. The appropriate comparison is therefore native audio versus transcript-only versus a fusion of both, stratified by what evidence the query requires.

8.2 Temporal sound and acoustic specificity

A sound recording can contain several events, and a query may depend on their order or duration. T-CLAP investigates temporal enhancement, while M2D-CLAP connects masked acoustic modeling with contrastive language–audio learning [115, 116]. A global vector may represent the overall acoustic setting while losing a brief event. Segment-level indexing or multi-vector audio representations can preserve this information at a storage cost.

Acoustic identity introduces another distinction. Two recordings of the same class of sound are semantically similar, but they need not rep-

resent the same source. A query for a specific speaker, instrument, or recording requires identity-sensitive features. The representation should not be judged only on broad class discrimination. Testing robustness to noise, recording devices, and compression also helps distinguish semantic transfer from reliance on incidental acoustic cues.

8.3 Music and cross-domain coverage

MuLan learns a joint representation of music audio and natural language, while weak language supervision and CLaMP explore complementary music representation settings [117–119]. Symbolic music, recorded performance, and textual description are not interchangeable inputs. A score contains pitch and rhythm structure but lacks recording timbre; a performance contains expressive and production details that may not appear in the score. A universal representation can connect these modalities while still requiring task-specific treatment of composition identity and rendition similarity.

GLAP addresses broader audio domains and languages [120]. The Massive Sound Embedding Benchmark, MSEB, brings a wider sound evaluation perspective, including work on benchmarking language models in that setting [97, 121]. These developments make it possible to ask whether a unified model retains acoustic discrimination while gaining text and visual compatibility. A text-heavy aggregate benchmark can otherwise hide weak non-speech audio performance.

8.4 Native audio in omni-modal systems

Gemini Embedding 2 reports native multimodal embedding with audio input, and Jina v5-omni includes audio within a geometry-preserving expansion strategy [13, 14]. Omni-Embed-Mini extends a fixed text space using caption-derived targets [66]. These routes differ in where the shared geometry is learned and how much of the audio encoder is adapted. Their common evaluation requirement is to distinguish transcript

semantics from acoustic information.

Caption-based distillation is particularly sensitive to this distinction. A caption that states “a person speaks” can align a recording to a broad concept, but cannot teach the student an omitted speaker identity or an exact prosodic pattern. Better captions can reduce this gap without eliminating it. A useful research direction is to combine semantic teachers with within-audio objectives that preserve distinctions missing from language, then test whether those distinctions remain accessible through an appropriate query interface.

9 Training design across heterogeneous tasks

9.1 Balancing modalities and tasks

A universal training mixture can contain many more text pairs than video or audio examples. Sampling in proportion to raw dataset size may prioritize the largest modality rather than the intended capability profile. Equal modality sampling has the opposite problem: it can oversample a small, noisy dataset. A principled mixture should specify target tasks, data quality, and retention constraints, then report the resulting exposure rather than only the total number of examples.

Modality-homogeneous batches can simplify negative construction because every query sees candidates of the same type. Mixed batches can teach calibration across candidate types but also introduce easy modality cues. Ovis-Embedding describes source-homogeneous batching as part of its training design [87]. Neither batching policy is universally preferable. The right experiment compares within-modal ranking, cross-modal transfer, and mixed-pool retrieval, where different policies may yield different strengths.

Loss weighting also matters. A weighted objective $\mathcal{L} = \sum_k \lambda_k \mathcal{L}_k$ should not be interpreted as balanced learning merely because the weights are equal. Tasks differ in example count, gradient scale, and difficulty. Reporting task-specific

progress and retention is more informative than assuming a scalar loss captures all capabilities. Adaptive weighting is a possible solution, but it should be evaluated for stability and for whether rare tasks receive enough useful updates.

9.2 Preserving existing capabilities

Continual modality expansion resembles a constrained adaptation problem. The new objective should improve a relation while limiting degradation on established ones. Freezing provides a hard constraint on selected parameters; distillation provides a softer constraint on outputs or pairwise relations. Jina v5-omni and Omni-Embed-Mini exemplify preservation-oriented designs [13, 66]. An unconstrained joint model may adapt more flexibly but requires explicit regression measurement.

A general objective can combine retrieval and retention terms:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \mathcal{L}_{\text{retrieval}} \\ & + \lambda_{\text{ret}} \mathbb{E}_x \|g_\theta(x) - g_0(x)\|_2^2 \\ & + \lambda_{\text{local}} \mathcal{L}_{\text{local}}. \end{aligned} \quad (5)$$

The expression summarizes the main objective terms used in this design space. The first term learns relevance, the second constrains an established space, and the third can preserve local or within-modal information. The terms may conflict: exact coordinate retention can limit adaptation, while excessive semantic compression can erase local distinctions. Ablation should therefore report the capability gained and the capability retained.

9.3 Dimension reduction and scalable outputs

Matryoshka Representation Learning trains nested prefixes of a vector to remain useful at different dimensions [122]. Related text work explores additional depth–dimension flexibility [123]. This is attractive for universal embeddings because storage and search costs grow with dimensionality. However, a prefix that preserves text retrieval may not preserve video or audio

retrieval equally well. Dimension sweeps should therefore be reported by modality and task.

Quantization likewise changes the geometry after training. A model can be accurate in full precision but lose fine distinctions under an index’s compression scheme. Evaluation should include the actual stored representation, not just the uncompressed embedding. The best training recipe depends partly on the intended retrieval engine: dense ANN, sparse inverted index, and multi-vector search impose different constraints. Training and indexing should be treated as a connected design problem.

9.4 Teacher quality, contamination, and supervision provenance

Synthetic data can increase coverage, but its provenance matters. A generator may have seen benchmark content during pretraining, and a filtering model may implicitly select examples similar to evaluation data. This does not establish contamination in any particular model; it motivates documenting source collections, generation prompts, and exclusion rules. Exact duplicate removal is useful but insufficient when paraphrases, resized images, or adjacent video clips remain.

Teacher selection should also be task-aware. A language-model teacher may judge semantic relevance well but be unreliable for small visual text, precise counts, or acoustic identity. Using several teachers can improve coverage while introducing disagreement. Rather than hiding disagreement with a single confidence threshold, a dataset can mark uncertain pairs and reserve them for evaluation or human review. This yields more interpretable supervision and reduces the risk of treating a plausible generated relation as ground truth.

10 Benchmarks and protocol-aware quantitative evidence

10.1 What benchmark families measure

Embedding evaluation spans retrieval, classification, clustering, semantic similarity, and related tasks. MTEB and MMTEB broaden text evaluation across tasks and languages [124, 125]. BEIR emphasizes heterogeneous retrieval and transfer [126]. MMEB, introduced with VLM2Vec, and its extensions expand the multimodal task set [8, 11, 127]. These suites expose more than one capability, but their aggregate scores remain dependent on task selection and weighting.

An average over datasets weights a small dataset and a large dataset equally under macro-averaging. An average over examples gives larger datasets greater influence. An average over task families makes a different choice again. None is inherently wrong, but the weighting determines what the headline score means. A model can improve an aggregate by gaining on many easy tasks while failing on a smaller, practically important group. Family-level and per-dataset results make these tradeoffs visible.

Specialized benchmarks test different weaknesses. COCO and Flickr30k provide image–text settings, while VATEX supports video–language evaluation [128–130]. SugarCrep and compositionality tests examine relational understanding, and negation studies expose failures concealed by topical matching [131–133]. BRIGHT evaluates reasoning-intensive retrieval [134]. MVEB focuses on visual identity [90]. A universal model should be assessed across these relation types rather than only on broad semantic matching.

10.2 Metrics are not interchangeable

Recall@ k measures retrieval of relevant items within a returned set, with the treatment of multiple positives defined by the benchmark. Mean reciprocal rank emphasizes the position of the first relevant result. NDCG accounts for

Table 3: Evaluation families and the capability each helps isolate. A strong result in one row does not establish the capabilities in another.

Family	Examples	Required interpretation
Text and multilingual	BEIR, MTEB, MMTEB	Check retrieval versus similarity, language coverage, and domain transfer.
Broad multimodal	MMEB, MMEB-V2, MMEB-V3	Record suite version, task composition, and modality-family scores.
Visual documents	ViDoRe V2, MMDocIR	Distinguish page relevance, document retrieval, and evidence localization.
Video and long video	VATEX, MMEB video tasks, LoVR	Specify direction, temporal unit, frame budget, and localization need.
Audio	MSEB, MAEB evaluations in reports	Separate speech content, non-speech semantics, and acoustic identity.
Difficult relations	SugarCrepe, FollowIR, BRIGHT, MVEB	Test composition, instructions, reasoning, and identity directly.
Long context	LongEmbed, MMLongEmbed	Measure relevant-detail retention, not just accepted input length.

graded relevance and position discounts:

$$\begin{aligned}
 \text{DCG}@k &= \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \\
 \text{NDCG}@k &= \frac{\text{DCG}@k}{\text{IDCG}@k}.
 \end{aligned} \tag{6}$$

Classification accuracy measures another property again. Similar numerical values do not make these metrics comparable. Retrieval direction also matters because candidate pools and positive sets differ. Image-to-text and text-to-image retrieval on the same corpus can yield different scores. The following tables preserve source-defined groupings and metrics rather than combine them into a leaderboard.

10.3 Broad image, video, and document comparison

Table 4 reproduces selected entries from Table 2 of the Qwen3-VL report, version 2 [85]. The report evaluates MMEB-V2 with 78 datasets and an updated out-of-distribution visual-document setting. Its re-evaluated baseline results should be interpreted within that protocol, not substituted for values from earlier original papers. Family scores show that overall improvement

can arise from very different gains across image, video, and visual-document tasks.

The 2B and 8B Qwen models provide a within-report scale comparison; the baseline rows compare different recipes and backbones. They do not isolate parameter count as the cause of improvement. Data, objectives, preprocessing, and task exposure also differ. A causal scaling claim would require controlled training across sizes with other variables held sufficiently constant.

10.4 Video and long-video evidence

Table 5 retains WAVE’s video and long-video breakdown from its Table 3 [12]. The MMEB-V2 video overall score combines different task families, whereas the long-video entries use different recall cutoffs. The breakdown preserves distinctions among classification, question answering, ordinary retrieval, moment retrieval, and theme-level long-video search.

Theme-to-clip Recall@25 cannot be compared directly with text-to-clip Recall@1: the task and retrieval budget both differ. Classification and question-answering aggregates also do not establish precise temporal localization. A universal video system needs evidence for whichever re-

Table 4: Selected MMEB-V2 results from Qwen3-VL Table 2, version 2 [85]. Values are source-reported aggregate scores on a 0–100 scale. Parameter labels follow that comparison.

Model	Image	Video	Visual document	All
VLM2Vec (2B)	59.7	28.6	44.0	47.7
VLM2Vec-V2 (2B)	64.9	34.6	69.2	59.2
GME (2B)	51.9	33.6	76.8	55.3
Qwen3-VL-Embedding (2B)	75.0	61.9	79.2	73.2
Qwen3-VL-Embedding (8B)	80.1	67.1	82.4	77.8

Table 5: WAVE-7B video results from Table 3, version 1 [12]. Scores use a 0–100 scale. Overall is the source-defined aggregate.

Evaluation	Metric or source grouping	Score
MMEB-V2 video	Overall	59.0
MMEB-V2 video	Classification aggregate	57.8
MMEB-V2 video	Question-answering aggregate	72.5
MMEB-V2 video	Retrieval, Recall@1	54.7
MMEB-V2 video	Moment retrieval, Recall@1	50.8
LoVR text-to-clip	Recall@1	62.9
LoVR theme-to-clip	Recall@25	66.0

Table 6: Selected Gemini Embedding 2 results from Table 1, version 1 [14]. Retrieval directions and metrics are retained.

Dataset and direction	Metric	Score
MSCOCO text → image	Recall@1	62.9
MSCOCO image → text	Recall@1	78.8
VATEX text → video	NDCG@10	68.8
MSR-VTT text → video	NDCG@10	68.0
ViDoRe V2	NDCG@10	64.9

trieval unit its deployment exposes.

10.5 Native multimodal retrieval and audio comparison

Gemini Embedding 2 reports retrieval across several modalities [14]. Table 6 selects image, video, and visual-document entries from its Table 1. These entries illustrate breadth; we do not combine them into a new aggregate or compare them with MMEB-V2 averages.

For MSEB audio passage retrieval, the same report’s Table 3 compares native audio with ASR-based input. Native audio achieves average MRR@10 of 73.99, with 75.58 for in-language and 72.56 for cross-language retrieval. The ASR-based comparison reports 70.40, 73.58, and 67.55 respectively [14]. This is evidence for that passage-retrieval setting, not superiority on every acoustic task. Non-speech recognition, source identity, and temporal event retrieval require separate tests.

Table 7: Jina embeddings v5-omni, Table 1, version 4 [13]. Modality columns are different benchmark-family aggregates. Parameters are total reported model sizes.

Variant	Parameters	Text	Image	Video	Audio	Average
Nano	0.95B	65.52	47.87	26.87	49.69	47.49
Small	1.57B	67.00	58.00	41.20	49.96	54.04

10.6 Geometry-preserving omni-modal extension

Jina v5-omni reports separate text, image, video, and audio evaluations [13]. Table 7 uses version 4 of the report. Text scores use MMTEB, image scores use MIEB, video scores use a nine-dataset MMEB subset, and audio scores use MAEB. The final average combines heterogeneous families; it is useful within this comparison but is not an MMEB-V2 score.

The breakdown shows why capacity comparisons need modality-specific evidence. A frozen-anchor model should be assessed on both established-task retention and new-modality effectiveness. Stable text weights establish one form of preservation, while cross-modal retrieval quality remains an empirical question.

10.7 Cross-paper conclusions and uncertainty

Tables 4–7 use different task mixes, metrics, families, and evaluation versions. Their overall scores are not ranked against one another. The supported synthesis is narrower: recent reports broaden coverage, expose substantial differences among modality families, and increasingly address preservation and efficiency. A definitive cross-model comparison requires a shared harness, frozen datasets, identical candidate pools, and aligned preprocessing budgets.

A small score difference may arise from sampling variation, approximate search, or a limited query set. Paired bootstrap intervals can quantify uncertainty for retrieval metrics; repeated runs can expose training variance. Per-query disagreement is useful when models have similar averages but succeed on different examples. We

recommend reporting uncertainty and disagreement rather than assuming an aggregate winner dominates every use case.

A minimum evaluation record identifies the dataset and version, split, direction, candidate pool, relevance labels, metric, instructions, pre-processing, checkpoint, dimension, and search configuration. For video and audio, duration and sampling are indispensable; for documents, resolution and OCR inclusion matter. We propose three layers of evidence: capability tests for requested relations, retention tests for established spaces, and operational tests for the deployed retrieval path.

11 Deployment and efficiency

11.1 End-to-end cost differs from parameter count

Retrieval has at least four costs: candidate pre-processing, candidate encoding and indexing, query encoding, and search or reranking. A video sampler or OCR stage can dominate runtime independently of the backbone. Closed APIs add batching and transfer overhead; local models add device-memory and concurrency constraints. Total and active parameters, media-token counts, and input duration should be reported separately.

For N dense vectors of dimension d with b bytes per coordinate, raw storage is

$$S_{\text{raw}} = Ndb. \quad (7)$$

As an arithmetic illustration, one billion 1,024-dimensional float32 vectors require 4.096×10^{12} bytes, or 4.096 decimal TB, before index structures, identifiers, metadata, or replicas. Multi-vector storage additionally scales with vectors

retained per item, making document and long-video indexing sensitive to granularity.

Query latency can be decomposed as

$$T_{\text{query}} = T_{\text{preprocess}} + T_{\text{encode}} + T_{\text{search}} + T_{\text{rerank}} + T_{\text{transfer}}. \quad (8)$$

Batching, hardware, input length, and cache behavior affect these terms. Encoder throughput omits much of the serving path. Tail latency is especially relevant when long and short inputs share a service.

11.2 Approximate search and hybrid interfaces

HNSW and GPU-based similarity-search systems provide foundations for scalable dense retrieval [135, 136]. Approximation introduces another recall boundary. An encoder evaluated with exact dot products can behave differently under a compressed or approximate index. Index settings should therefore be tuned and evaluated with the representation, retaining exact-search results as a reference.

Sparse outputs support inverted-index retrieval and lexical evidence; dense outputs support semantic matching. M3-Embedding combines multiple text retrieval functions, while UEmbed extends sparse-dense unification to multimodal decoder representations [25, 137]. Hybrid search needs a fusion policy: raw sparse and dense scores should not be summed without examining their scale and calibration. Rank-based fusion avoids some scale issues but still requires task-specific validation.

Multi-vector search offers a third interface. Compression, pruning, and specialized engines reduce its cost [21, 22, 138]. Its benefit is strongest when local evidence changes relevance: rare document terms, selected regions, or brief audiovisual events. A single universal vector and a universal retrieval system are different goals; the latter may expose coordinated representation types.

11.3 Reranking and evidence selection

Rerankers compare a query with a candidate through richer interaction than a stored-vector score. Qwen3-VL’s framework illustrates this division [85]. A pipeline should report first-stage recall at the reranking depth, final quality, and total latency. Larger shortlists can improve results but change the budget. A strong reranker cannot recover a candidate that first-stage search excluded.

In retrieval-augmented generation, supported answering is often the final objective. VisRAG provides a visual-document example [93]. A model can retrieve a plausible page while failing to read the required cell; a generator can also produce a correct-looking answer from irrelevant evidence. Relevance, evidence coverage, and answer support should be measured separately to locate errors in encoding, selection, localization, or generation.

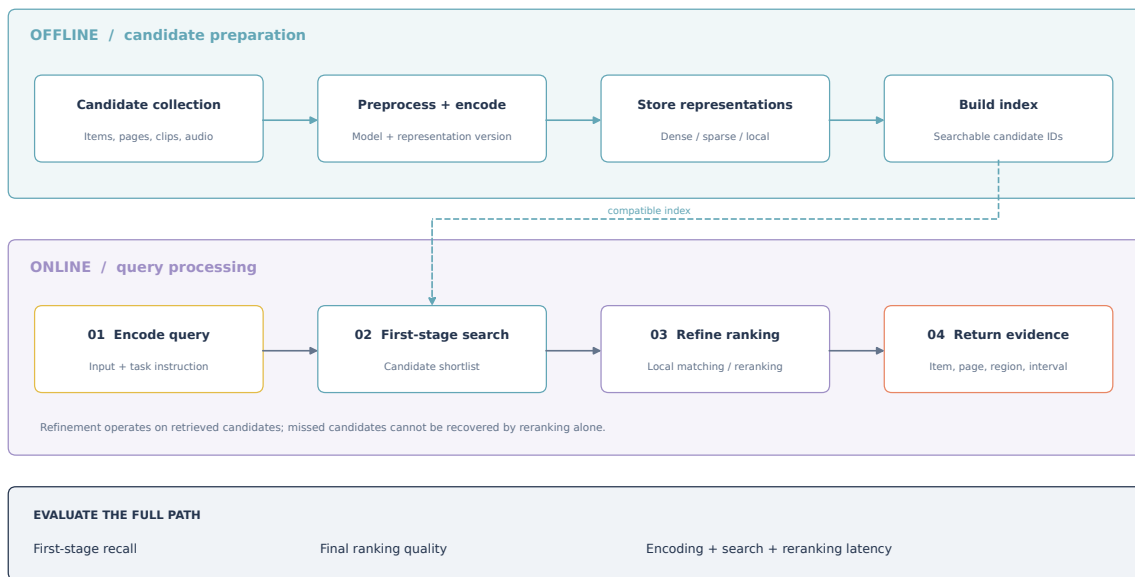
11.4 Index evolution and compatibility

Updating an encoder can make old and new vectors incompatible despite identical dimensions. Mixing checkpoints in one index creates a shift invisible to evaluation using freshly encoded candidates. Migration requires re-encoding, a validated transformation, or versioned indexes. The cost can outweigh a small benchmark gain.

Frozen-anchor expansion can reduce migration burden for text, but new modalities still require ingestion and calibration. A compatibility test compares new queries directly with old stored vectors. It should also check duplicate handling, score distributions, and exposure across candidate types. Preserved weights are operationally valuable when these tests confirm actual index reuse.

From multimodal inputs to retrieved evidence

Offline candidate preparation and online query processing share a compatible representation interface.



Schematic workflow: index and refinement choices depend on representation type and retrieval task.

Figure 6: A multimodal retrieval pipeline separates offline candidate encoding from online query processing. Later interaction improves ranking only among retrieved candidates, so first-stage recall and end-to-end cost remain essential evaluation quantities.

12 Failure modes and research agenda

12.1 Composition, negation, and relation specificity

Vision–language similarity can overuse object names while underusing relations and order [131, 132]. Negation is a direct test: an excluded property may still increase similarity because its name occurs in the query [133]. Universal training does not automatically solve this problem if the supervision continues to reward broad matching.

Evaluation should use minimally different queries and candidates: the same objects in different relations, actions in reversed order, or descriptions with negated constraints. Controlled examples isolate mechanisms; natural examples provide realism. Agreement across both is stronger than a single aggregate. Theoretical

limits of fixed embedding retrieval further motivate matching representation capacity and interaction to the relevance relation [139]. Some tasks may require local vectors or query-conditioned scoring.

12.2 Identity, temporal evidence, and missing modalities

MVEB brings visual identity into universal embedding evaluation [90]. Semantic training places instances of a class together, while identity retrieval must distinguish them. Conditioning, factorized features, or complementary heads may support both. Evaluation needs same-class negatives and changes in viewpoint, lighting, and background, with the intended identity relation explicitly defined.

Long audiovisual retrieval requires retention of rare events. The indexing process cannot know every future query, so global summaries

need complementary local representations. Tests should separate event visibility under sampling from representation failure after the event is visible. Missing, corrupted, or mismatched streams also deserve paired evaluation. Strong fused-input results alone do not establish robustness to an absent modality.

12.3 Multilingual and domain transfer

Coverage should include scripts, dialects, cultural concepts, and domain vocabulary. Multilingual benchmarks and image-text models provide foundations [46, 47, 125], but task-specific transfer remains necessary. A language can retrieve common visual concepts well while failing on specialized documents or culturally specific scenes.

Scientific figures, industrial sounds, and instructional videos differ from web captions. Domain adaptation can improve them while damaging a general space. Retention-aware training and versioned indexes offer a path if in-domain gains and out-of-domain losses are both measured. Universal claims should make these boundaries visible.

12.4 Reproducibility and operational universality

Reproducible evaluation requires preprocessing, instructions, dataset versions, candidate pools, and scoring code in addition to a checkpoint. Revised reports should identify changed settings. Training-cost disclosure should include hardware, accelerator-hours, precision, sequence sizes, data passes, and teacher or preprocessing costs. Parameter count alone is an incomplete proxy; our selected result tables make no training-cost ranking.

Five problems deserve particular attention: separating relevance relations, preserving compatibility with old indexes, calibrating mixed candidate pools, matching representations to serving budgets, and documenting data and preprocessing. Together they determine whether

a broadly capable encoder remains useful in a working retrieval system.

Promising directions include factorized shared and modality-specific features, task-conditioned scoring, preservation-constrained expansion, and adaptive granularity. Each needs a matched cost budget: extra interaction or capacity can improve accuracy while making large-scale retrieval impractical. A capability profile with retention and systems evidence offers a clearer account of progress than an opaque aggregate.

13 Conclusion

Multimodal representation learning now offers several routes to a common retrieval interface. Contrastive towers remain efficient for paired matching; anchor-based binding reduces the need for all-pair supervision; multimodal language-model backbones support mixed inputs and task instructions. Sparse and local representations extend this interface when a single vector loses information that matters for ranking.

The strongest design depends on the evidence a task needs. Document retrieval must preserve text and layout, video retrieval must retain the relevant event and its order, and audio retrieval must distinguish linguistic content from acoustic detail. Broader modality coverage is valuable, but it cannot replace tests of these capabilities. The reported comparisons also show that benchmark averages depend heavily on the tasks and protocols included.

Further progress will require closer integration of training, evaluation, and indexing. Models should preserve established capabilities as new modalities are added, remain calibrated in mixed candidate pools, and retain their gains under the compression and approximation used in deployment. These requirements provide a practical basis for judging unified representations beyond a headline score.

Table 8: Open problems and diagnostic tests suggested by the preceding analysis.

Problem	Failure mechanism	Diagnostic
Instruction insensitivity	Content dominates requested relation	Contrasting instructions with fixed candidates.
Identity loss	Semantic learning erases instance differences	Same-class negatives and viewpoint changes.
Temporal omission	Sampling or pooling loses brief events	Visibility rate, order tests, interval recall.
Modality preference	Mixed-pool scores are miscalibrated	Candidate-type exposure and per-type recall.
Index drift	Adaptation changes established coordinates	Old-index/new-query tests and retention suites.
Long-context dilution	Pooling suppresses small relevant details	Distractor-length sweeps and evidence coverage.
Serving mismatch	ANN or compression changes ranking	Exact versus deployed-index accuracy and latency.

A Reporting checklist

1. **Define relevance and retrieval units.** Specify semantics, identity, evidence, composition, or temporal relations and accepted positives.
2. **Specify inputs.** Record modalities, mixed formats, token limits, image resolution, page policy, frame sampling, and audio duration.
3. **Describe representations.** Report pooling, dimensions, normalization, sparse or local outputs, instructions, and index reuse across tasks.
4. **Document training.** Describe data provenance, synthesis, negatives, mixture exposure, retention objectives, teachers, and available compute.
5. **Freeze evaluation.** Identify versions, splits, candidate pools, directions, metrics, and preprocessing. Report family scores and uncertainty.
6. **Test retention and compatibility.** Evaluate established tasks and new queries against old stored vectors where reuse is claimed.
7. **Measure serving.** Report encoding, indexing, memory, query and tail latency, compression, ANN settings, and reranking

depth.

8. **Audit difficult cases.** Include instruction contrasts, negation, identity, order, long-context distractors, and modality ablations.

B Quantitative evidence notes

Tables 4–7 retain source-defined aggregates and do not recompute them from rounded values. Qwen3-VL Table 2 uses its own MMEB-V2 baseline re-evaluation. WAVE Table 3 separates video families and LoVR tasks. Gemini Table 1 mixes Recall@1 and NDCG@10; its Table 3 supplies the audio passage-retrieval comparison. Jina v5-omni Table 1, version 4, combines distinct benchmark families. No common ranking is implied.

The bibliography identifies the primary archive versions used for these comparisons. Subsequent benchmark or report revisions should be checked against those versions before their values are combined.

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